

## IGCSE Physics 0625 — Equation &amp; Data Sheet

Cambridge IGCSE 0625 (2023-2027 syllabus). Every equation with quantities, units and the topic it appears in.

## MOTION &amp; FORCES

|                     |                                                                                    |
|---------------------|------------------------------------------------------------------------------------|
| Speed               | $v = d / t$ $v$ : speed (m/s), $d$ : distance (m), $t$ : time (s)                  |
| Acceleration        | $a = (v - u) / t$ $a$ : (m/s <sup>2</sup> ), $v$ : final, $u$ : initial velocity   |
| Equations of motion | $v^2 = u^2 + 2as$ ; $s = ut + (1/2) a t^2$ (uniform $a$ )                          |
| Force               | $F = m a$ $F$ : force (N), $m$ : mass (kg), $a$ : acceleration                     |
| Weight              | $W = m g$ $g = 9.8 \text{ N/kg}$ (use 10 unless told)                              |
| Momentum            | $p = m v$ $p$ : momentum (kg m/s)                                                  |
| Impulse             | impulse = $F t$ = change in momentum                                               |
| Moments             | moment = $F \times d$ ( $d$ : perpendicular distance from pivot)                   |
| Hooke's law         | $F = k x$ $k$ : spring constant (N/m), $x$ : extension (m)                         |
| Pressure            | $p = F / A$ $p$ : pressure (Pa or N/m <sup>2</sup> ), $A$ : area (m <sup>2</sup> ) |

## ENERGY &amp; WORK

|                |                                                                      |
|----------------|----------------------------------------------------------------------|
| Work done      | $W = F d$ $W$ : work / energy (J), $d$ : distance moved by force (m) |
| Kinetic energy | $KE = (1/2) m v^2$                                                   |
| GPE            | $GPE = m g h$                                                        |
| Power          | $P = W / t = E / t$ $P$ : power (W)                                  |
| Efficiency     | efficiency = useful output / total input (x 100%)                    |

## THERMAL PHYSICS

|                        |                                                                            |
|------------------------|----------------------------------------------------------------------------|
| Specific heat capacity | $E = m c (\Delta \theta)$ $c$ : (J/(kg C)), $\Delta \theta$ : temp rise    |
| Specific latent heat   | $E = m L$ $L$ : (J/kg) — fusion or vaporisation                            |
| Pressure & volume      | $p V = \text{constant}$ (fixed mass of gas, constant temperature)          |
| Pressure & temperature | $p / T = \text{constant}$ (constant volume, $T$ in kelvin: $K = C + 273$ ) |

## WAVES &amp; OPTICS

|                  |                                                                                      |
|------------------|--------------------------------------------------------------------------------------|
| Wave speed       | $v = f (\lambda)$ $f$ : frequency (Hz), $\lambda$ : wavelength (m)                   |
| Refractive index | $n = \sin i / \sin r = (\text{speed of light in vacuum}) / (\text{speed in medium})$ |
| Critical angle   | $\sin C = 1 / n$                                                                     |
| Magnification    | magnification = image height / object height                                         |

## ELECTRICITY &amp; MAGNETISM

|                     |                                                                           |
|---------------------|---------------------------------------------------------------------------|
| Charge              | $Q = I t$ $Q$ : charge (C), $I$ : current (A), $t$ : time (s)             |
| Voltage             | $V = W / Q$ $V$ : voltage (V) = energy per unit charge (J/C)              |
| Ohm's law           | $V = I R$ $R$ : resistance (ohm)                                          |
| Electrical power    | $P = V I = I^2 R = V^2 / R$                                               |
| Energy transferred  | $E = V I t$                                                               |
| Series / parallel R | series: $R = R_1 + R_2 + \dots$ ; parallel: $1/R = 1/R_1 + 1/R_2 + \dots$ |
| Transformer         | $V_p / V_s = N_p / N_s$ (ideal: power in = power out)                     |

## ATOMIC PHYSICS / RADIOACTIVITY (no equations — know these)

|           |                                                              |
|-----------|--------------------------------------------------------------|
| Half-life | time for half the unstable nuclei to decay (read from graph) |
|-----------|--------------------------------------------------------------|

**Background count**

always subtract background from measured count rate

**UNITS & PREFIXES****Base units**

m, kg, s, A, K

**Prefixes** $n = 10^{-9}$ ; micro =  $10^{-6}$ ; m =  $10^{-3}$ ; k =  $10^3$ ; M =  $10^6$ ; G =  $10^9$ **Converting** $1 \text{ m}^2 = 10^4 \text{ cm}^2$ ;  $1 \text{ m}^3 = 10^6 \text{ cm}^3$ ;  $1 \text{ g} = 10^{-3} \text{ kg}$